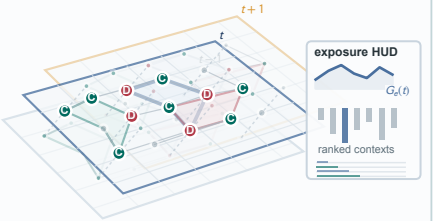
a Dense temporal hypergraph layer

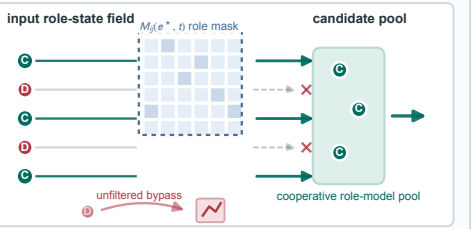
layered temporal stack with group-level exposure signals



Temporal hyperedges encode group contexts, not just pairwise contacts.

b Matrix role filter / spatial masking gate

strategy state vectors are gated by cooperation-conditioned eligibility



endogenous feedback

c Continuous signal flow and mathematical pipeline



1 Observe local group

$$P_i(t) = \sum_{e \ni i} [r\, n_C(e,t) - s_i(t)]$$

2 Sample context

$$G_e(t) = q_e(t) \max\{0,\, r\, n_C(e,t) - 1\}$$

3 Screen roles

$$C_i(e^*, t) = \{j \in e^* : s_j(t) = 1,\ j \neq i\}$$

4 Update strategy

$$\Pr(i \leftarrow j) = [1 + \exp\{-\beta \Delta P\}]^{-1}$$
$$\Delta P = P_j - P_i$$

Successful-group context selects where the signal is sampled; role screening controls who transmits.